Lunar Remote Sensing Image Matching under a Limited Computational Budget

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Abstract. Lunar remote sensing image matching faces strict computational constraints in practical applications. This work investigates the trade-off among matching success rate, geometric accuracy, and computational cost under limited budgets. A unified evaluation framework is constructed using 60 multi-illumination Chang'e-3 image pairs. Input conditions, candidate matching, geometric verification, and success criteria are kept consistent throughout the experiments. The influence of RANSAC sampling limits and reprojection thresholds is first evaluated. The results indicate that increasing the sampling limit alone does not significantly improve the success rate. In contrast, the reprojection threshold directly affects the success rate, inlier ratio, and geometric error. ORB, SIFT, fixed-parameter ALIKE, and two adaptive variants are then compared within the same framework. On this basis, a budget-adaptive multi-scale matching method, termed BAMM-ALIKE, is proposed. The method allocates computational resources by jointly adjusting image scale and the number of keypoints. Feasible configurations are selected by using geometric verification feedback under different time budgets. Experimental results demonstrate that BAMM-ALIKE maintains stable performance across budget levels. It further achieves a more favourable balance between accuracy and efficiency, especially under medium-budget conditions, which supports robust image matching in resource-constrained lunar remote sensing tasks.

1 Introduction

Lunar remote sensing image matching is a basic step in orbital image mosaicking and mapping tasks [1]. Lunar surface images are affected by variations in illumination and uneven texture distribution. As a result, matching results can fluctuate, disrupting mosaicking and mapping [1]. Real engineering systems also face delay and computing limits. High resolution and complex inference are hard to process in real time. A fixed low-resource setting is faster. However, it can distort the relationship between stitching and localisation in difficult image pairs. This may cause map misalignment or structural deformation. It also reduces the accuracy and reliability of remote sensing images.

Learning-based matching methods usually have stronger representation ability than traditional local features. They can improve matching usability on difficult samples [1]. For example, SuperPoint, SuperGlue, and LoFTR improve matching quality in terms of

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keypoints and descriptors, matching inference, and detector-free association [2-4]. However, these methods usually require more computation. They are therefore more restricted during deployment. To reduce cost, recent studies have emphasised the trade-off between accuracy and efficiency. The lightweight front-end ALIKE shows advantages in both accuracy and speed [5]. LightGlue greatly accelerates matching through adaptive inference and pruning [6]. Conditional computation emphasises on-demand allocation of computation under resource limits [7].

However, most existing studies do not provide a clear closed-loop resource scheduling mechanism for scenes in which sample difficulty changes over time. It is therefore hard to guarantee stable outputs for inputs of different difficulty under fixed resources. To address this issue, this paper proposes BAMM, an adaptive lunar image-matching framework under limited computational constraints. It uses the lightweight learning-based front-end ALIKE [5]. It treats input scale and keypoint resources as allocable computation. It also uses the inlier ratio and reprojection error from RANSAC-based geometric verification as feedback signals [8,9]. In this way, the resource setting can be selected and scheduled adaptively under a limited computational budget [8,9]. Under a unified success criterion, this paper uses running time as an observable indicator of computational cost. It evaluates the effects of geometric verification parameters on stability and accuracy. ORB and SIFT are used as traditional baselines [10,11]. The experiments further verify the accuracy gain of BAMM-ALIKE under a limited computational budget and its diminishing-return behaviour.

2 Methods

2.1 Data

2.1.1 Data source

The experiments in this paper use 60 image pairs captured by the Chang'e-3 lunar camera [1]. Chang'e-3 images are selected for three reasons. First, the dataset naturally contains imaging differences for the same area under different solar elevation and azimuth angles. As shown in Fig. 1, the shadowed regions at the same location differ markedly over time. This makes the dataset suitable for testing the stability of a matching algorithm when input difficulty changes. Second, Chang'e-3 images have a clear engineering background and imaging chain. They are therefore closer to orbital image mosaicking and mapping applications. Third, the dataset is moderate in size. This makes it convenient to conduct ablation experiments and evaluate budget curves within a unified pipeline and a criterion. It also highlights the main theme of controllability under a limited computational budget.

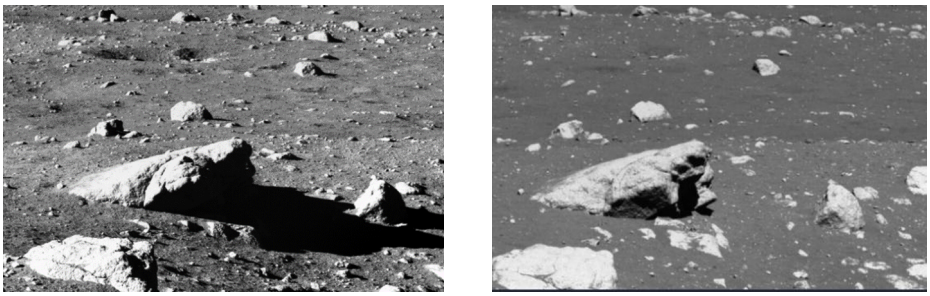


Fig. 1. Partial original lunar surface images captured by Chang'e-3 [1]

2.1.2 Data preprocessing

To unify the input format and control computational cost, this paper applies three preprocessing steps to the Chang'e-3 images. The first step is to convert to 16-bit grayscale. If an original image is in colour, it is converted to an 8-bit grayscale image. The 8-bit grayscale image is then linearly mapped to a 16-bit dynamic range. Each pixel value is multiplied by 257. In this way, 0–255 is mapped to 0–65535. The result is saved as a 16-bit grayscale image for later batch processing and management.

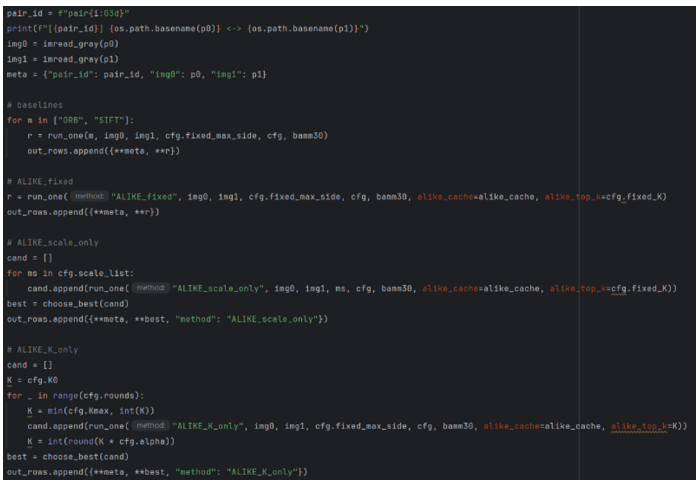
The second step screens candidate image pairs by similarity. After a 16-bit grayscale image is read, its brightness is first linearly stretched to a unified range and converted to an 8-bit grayscale image. Local contrast enhancement is then applied. An 8×8 block scheme is used, and the maximum enhancement factor is set to 2.0. Finally, downsampling is performed only when the original image is larger. The maximum image side length is limited to 512 pixels. This step is used only to compute the similarity score. It does not change the input for the main evaluation stage.

The third step unifies scale in the main evaluation stage. All methods run on grayscale images. The images are resized proportionally according to a preset maximum side length. For example, the maximum side length can be set to 2000 pixels. This keeps scale consistent and controls computational cost.

2.2 Experimental steps

2.2.1 Compared methods and unified evaluation pipeline

To ensure reproducibility and comparability, all methods use unified input conditions, candidate-match generation, the same geometric verification model, and the same success criterion. The compared methods include ORB, SIFT, ALIKE_fixed, ALIKE_scale_only, ALIKE_K_only, and BAMM-ALIKE [10,11]. ORB and SIFT are used as traditional baselines [10,11], and both are implemented with OpenCV [12]. The ALIKE-based methods use the alike-n model [5]. Only the input scale and the keypoint resource setting are changed in different experiments. The network structure is not changed. The inference device is CUDA. The keypoint score threshold (scores_th) is 0.05. The detection upper limit (n_limit) is 12000. The main loop implementation is shown in Fig. 2. Under these unified settings, performance differences primarily arise from feature-extraction and resource-allocation strategies rather than from inconsistent back-end evaluation conditions.



```
pair_id = f"pair{i:03d}"
print(f"[{pair_id}] {os.path.basename(p0)} <-> {os.path.basename(p1)}")
img0 = imread_gray(p0)
img1 = imread_gray(p1)
meta = {"pair_id": pair_id, "img0": p0, "img1": p1}

# baselines
for m in ["ORB", "SIFT"]:
    r = run_one(m, img0, img1, cfg.fixed_max_side, cfg, bamm30)
    out_rows.append({**meta, **r})

# ALIKE_fixed
r = run_one( method: "ALIKE_fixed", img0, img1, cfg.fixed_max_side, cfg, bamm30, alike_cache=alike_cache, alike_top_k=cfg.fixed_K)
out_rows.append({**meta, **r})

# ALIKE_scale_only
cand = []
for ms in cfg.scale_list:
    cand.append(run_one( method: "ALIKE_scale_only", img0, img1, ms, cfg, bamm30, alike_cache=alike_cache, alike_top_k=cfg.fixed_K))
best = choose_best(cand)
out_rows.append({**meta, **best, "method": "ALIKE_scale_only"})

# ALIKE_K_only
cand = []
K = cfg.K0
for _ in range(cfg.rounds):
    K = min(cfg.Kmax, int(K))
    cand.append(run_one( method: "ALIKE_K_only", img0, img1, cfg.fixed_max_side, cfg, bamm30, alike_cache=alike_cache, alike_top_k=K))
    K = int(round(K * cfg.alpha))
best = choose_best(cand)
out_rows.append({**meta, **best, "method": "ALIKE_K_only"})
```

Fig. 2. Main loop implementation of different comparison methods under the unified evaluation framework (Picture credit: original)

2.2.2 Candidate-match generation and geometric verification

All methods use the same back-end for candidate matching and geometric verification. In the candidate-matching stage, ORB and SIFT use KNN nearest-neighbour search with k = 2 [11]. Lowe's ratio test is then used to remove ambiguous matches. The ALIKE branch calls a unified descriptor-matching function. It also uses nearest-neighbour search (k = 2) and the same ratio-test strategy [11]. The ratio threshold is set to 0.90.

In the geometric verification stage, RANSAC is used to estimate the homography matrix H [8,9]. In the experiments, the reprojection error threshold (thr) is set to 3.0 px. The target confidence (p0) is 0.99. The maximum number of iterations (Nmax) is 5000. Let M be the number of candidate matches, and let I be the number of RANSAC inliers. The inlier ratio is defined as

$$w = \frac{I}{M}. \tag{1}$$

When the minimum sample size of the homography model is 4, the number of samples required to reach p0 can be written as

$$N = \frac{ln(1-p_0)}{ln(1-w^s)}. \tag{2}$$

Here, N denotes the theoretical number of samples required to draw an all-inlier sample at least once under the current inlier ratio.

2.2.3 Success criterion and evaluation metrics

To avoid judging matching quality based on a single metric, this paper uses a composite success criterion. The script first performs a basic check based on the number of inliers and the theoretical number of samples. Specifically, the inlier number I must satisfy $I \geq I_{min}$ (the minimum inlier number), and the theoretical sample number N must satisfy $N \leq N_{max}$ On this basis, geometric consistency is further constrained by geometric error and the inlier ratio. In the experiments, I_{min} = 8, and the minimum inlier ratio (inlier_ratio_min) is 0.0. The median reprojection error threshold (geom_med_max) is set to 2.4 px by default. The RMSE threshold (geom_rmse_max) is set to 3.6 px by default. Therefore, a match is defined as successful only when the candidate passes RANSAC verification and simultaneously

satisfies the minimum inlier number, the sampling upper limit, the upper bound on median reprojection error, the RMSE upper bound, and the minimum inlier-ratio constraint. On this basis, the following metrics are reported uniformly. Success rate: the ratio of image pairs that satisfy the success criterion to the total number of image pairs. RMSE: the root mean square of the inlier reprojection errors in successful samples, in pixels. Inlier ratio: the ratio of the number of inliers to the total number of candidate matches. Runtime: the end-to-end time for a single image pair to complete scaling, feature extraction, matching, and geometric verification, in milliseconds.

2.2.4 BAMM: Budget-Adaptive Multi-scale Matching

The goal of BAMM is to automatically select a computational-resource configuration for each image pair under a given time budget B (ms). The aim is to make geometric verification succeed as stably as possible while reducing geometric error. This paper treats two variables that directly affect computational cost as allocable resources: processing scale, i.e., the maximum side length after image resizing, and keypoint resources, i.e., the maximum number of keypoints used in matching. BAMM uses the inlier ratio and reprojection error produced by geometric verification as feedback signals. It also uses end-to-end running time as the measure of computational cost in milliseconds. In this way, the computational budget is tied to a reproducible engineering metric.

2.2.5 Ablation settings and BAMM scheduling settings

To analyse the effects of different resource variables on matching performance, this paper designs fixed-configuration experiments, single-variable ablation experiments, and BAMM joint-scheduling experiments. In the fixed-configuration experiment, ALIKE_fixed uses a unified maximum input side length (fixed_max_side = 2000 px) and a maximum number of output keypoints (fixed_K = 2048). In the single-variable ablation experiments, ALIKE_scale_only fixes the keypoint upper limit (K) at 2048 and runs only at four input scales: 800, 1200, 1600, and 2000 px. ALIKE_K_only fixes the maximum input side length at 2000 px and changes only the number of keypoints. It starts from K0 = 1024 and increases by a factor of alpha = 2 for 4 rounds. The maximum number of keypoints (Kmax) is 8192. These two ablation settings allow the independent effects of input scale and keypoint resources on matching performance to be evaluated separately.

On this basis, BAMM-ALIKE schedules input scale and keypoint resources simultaneously. The core scheduling process is shown in Fig. 3. For each image pair, it first traverses the candidate scale set of 800, 1200, 1600, and 2000 pixels. It then increases the keypoint upper limit from K0 = 1024 at each scale. This forms multiple candidate configurations of (scale, K). For each configuration, ALIKE feature extraction, candidate matching, and RANSAC geometric verification are executed completely once. The resulting values of time, RMSE, and inlier_ratio are recorded. If the end-to-end running time of a configuration exceeds the computational budget, that configuration is removed directly. Among the configurations that satisfy the budget, those that satisfy the unified success criterion are retained first. The final output is then determined based on a smaller geometric error and better inlier performance. The experimental results for BAMM-ALIKE are reported for three budget levels: B = 1500, 2500, and 4000 ms. These results are used to analyse the trade-offs among success rate, accuracy, and running time under varying computational budgets.

```
# BAMM budgets (ms)
for B in cfg.budget_ms_list:
    cand = []
    for ms in cfg.scale_list:
        K = cfg.K0
        for _ in range(cfg.rounds):
            K = min(cfg.Kmax, int(K))
            r0 = run_one( method: "BAMM_ALIKE", img0, img1, ms, cfg, bamm30, alike_cache=alike_cache, alike_top_k=K)
            if r0["time_ms"] <= float(B):
                r0["budget_ms"] = int(B)
                r0["chosen_max_side"] = int(ms)
                r0["chosen_K"] = int(K)
                cand.append(r0)
            K = int(round(K * cfg.alpha))
    if len(cand) == 0:
        out_rows.append({**meta, "method": f"BAMM_ALIKE_B{B}", "budget_ms": int(B), "chosen_max_side": None, "chosen_K"
                         "max_side": None, "K": None, "M": 0, "I": 0, "w_hat": 0.0, "N_required": float("inf"),
                         "geom_median_px": float("inf"), "geom_rmse_px": float("inf"), "inlier_ratio": 0.0,
                         "success": 0, "success_geom": 0, "time_ms": float("inf"), "scale0": None, "scale1": None})
    else:
        best = choose_best(cand)
        out_rows.append({**meta, **best, "method": f"BAMM_ALIKE_B{B}"})
```

Fig. 3. Main scheduling loop of BAMM-ALIKE under computational budget constraints (Picture credit: original)

Table 1. Experimental parameter settings

Parameter	Value	Role
k	2	Number of nearest neighbours in KNN search
ratio	0.9	Threshold of the KNN ratio test
thr	3.0 px	RANSAC reprojection error threshold
p0	0.99	Target confidence of RANSAC
Nmax	5000	Maximum number of RANSAC iterations
s	4	Minimum sample size of the homography model
I_{min}	8	Minimum inlier threshold for a successful match
geom_med_max	2.4 px	Upper bound of median reprojection error
geom_rmse_max	3.6 px	Upper bound of RMSE
fixed_max_side (px)	2000 px	Maximum input side length in the fixed configuration
fixed_K	2048	Keypoint resource in the fixed configuration
K0	1024	Initial value in keypoint search
Kmax	8192	Upper bound in keypoint search
α	2	Multiplicative increment of keypoints
rounds	4	Number of keypoint-increment rounds

Table 1 lists only the key parameters that affect the reproduction of the experimental results. Data preprocessing, model implementation, and result output have been described in the main text.

3 Results

3.1 Effect of geometric verification parameters

Under the same dataset and a fixed feature-matching pipeline, this study compares four groups of RANSAC parameter settings under different computational budgets and thresholds (Table 3). The main differences among the four settings lie in the random-sampling upper limit (Nmax) and the inlier threshold (thr) (Table 2).

Table 2. Parameter settings of the experimental groups

Experiment ID	Nmax	thr/px
1	1.0×10^9	8
2	5000	5
3	5000	3
4	5000	8

Table 3. Matching statistics of the four groups on Chang'e-3 images

Parameters	1	2	3	4
Success rate	0.333333	0.333333	0.7	0.6
Mean RMSE	0.100302	0.04976	0.702203	1.520008
Median RMSE	0.102544	0.039091	0.489624	1.450959
Mean geometric error	0.002127	0.001318	0.128801	0.632492
Inlier ratio	0.903458	0.924286	0.731703	0.618211
Median inlier count	6621.15	6892.2	2829.357	3341.25
Mean time (s)	5.662852	5.552181	5.519825	5.642468
Median time (s)	5.695791	5.6185	5.457383	5.544639

3.1.1 Comparison of success rate

Among settings with a random-sampling upper limit of 5000, the success rate is highest when the geometric threshold is set to 3 pixels, at about 0.7. In contrast, the setting with a random-sampling upper limit increased to 10^9, and the geometric threshold was kept at 8 pixels, giving a success rate of about 0.33.

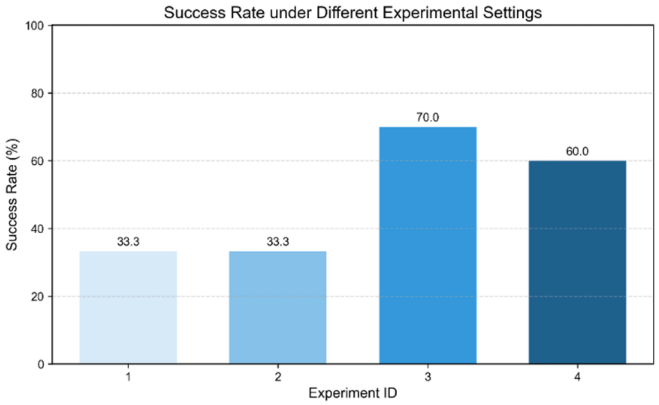
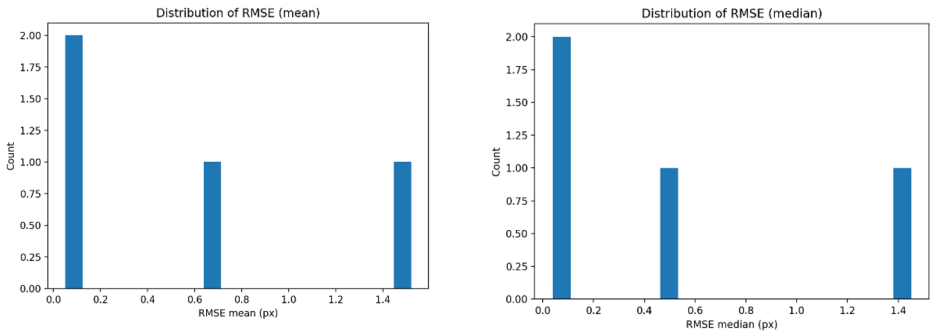


Fig. 4. Success rate comparison across the four experimental groups (Picture credit: original)

3.1.2 Distribution of RMSE for successful samples

Among settings with a limited iteration upper bound, the best accuracy is achieved when the geometric threshold is set to 5 pixels. The median RMSE of successful samples is about 0.039, and the mean is about 0.05. When the geometric threshold is set to 3 pixels, the success rate is highest, but the RMSE of the successful samples increases substantially, with a mean of about 0.70. When the geometric threshold is 8 pixels, the success rate remains relatively high, but the error increases further, with a mean of about 1.52. For the control setting with an extremely large iteration upper limit and an 8-pixel geometric threshold, the accuracy is better than that of the limited-iteration setting with the same threshold. However, it does not increase the success rate.



(a) Distribution of the mean RMSE of successful samples

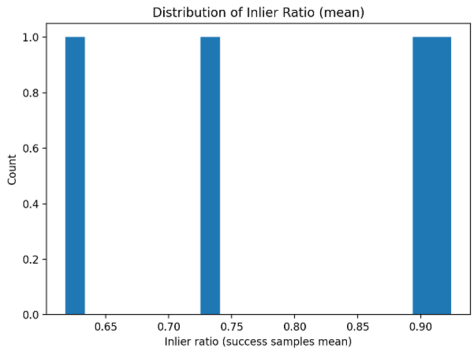
(b) Distribution of the median RMSE of successful samples

Fig. 5. Distribution of RMSE statistics for successful samples (Picture credit: original)

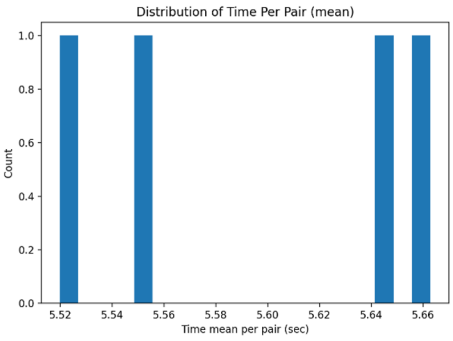
3.1.3 Distribution of inlier ratio and average runtime

Among the settings with a limited iteration upper bound, the highest mean inlier ratio for successful samples appears when the geometric threshold is 5 pixels, at about 0.92. The lowest mean inlier ratio occurs at a geometric threshold of 8 pixels, at about 0.62. The control

group has a mean inlier ratio of about 0.90 for successful samples. The average running times of the four experiments are all concentrated between about 5.5 and 5.7 seconds. The presence or absence of an iteration upper limit does not lead to a significant difference in average time consumption.



(a) Distribution of inlier ratio for the four experimental groups



(b) Distribution of average running time

Fig. 6. Distribution of inlier ratio and average running time for the four experimental groups (Picture credit: original)

3.2 Multi-method comparison

Using 60 Chang'e-3 multi-illumination image pairs, this paper evaluates ORB, SIFT, fixed-parameter ALIKE (ALIKE_fixed), keypoint-only adaptation (ALIKE_K_only), scale-only adaptation (ALIKE_scale_only), and BAMM under a limited computational budget (B = 1500/2500/4000 ms) in a unified way. Table 4 summarises the geometric success rate, RMSE of successful samples, inlier ratio, and runtime of each method.

Table 4. Matching statistics of different methods on Chang'e-3 images

Method	success rate	RMSE (px)	inlier ratio	runtime (ms)	successful pairs
ORB	0.8	0.8779	0.707	138.72	48
SIFT	0.8667	0.3727	0.7167	833.18	52
ALIKE_fixed	0.7667	0.8099	0.689	2519.45	46
ALIKE_K_only	0.7833	0.5671	0.7356	2544.03	47
ALIKE_scale_only	1	0.4619	0.7918	676.23	60
BAMM_ALIKE_B1500	1	0.4123	0.7396	614.15	60
BAMM_ALIKE_B2500	1	0.377	0.7706	907.59	60
BAMM_ALIKE_B4000	1	0.3767	0.7704	1054.17	60

3.2.1 Comparison of matching statistics across methods

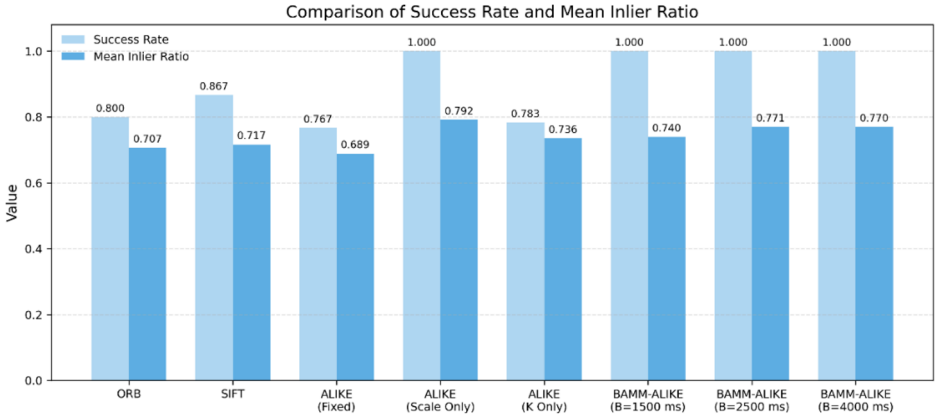


Fig. 7. Success rate and inlier ratio comparison among different methods (Picture credit: original)

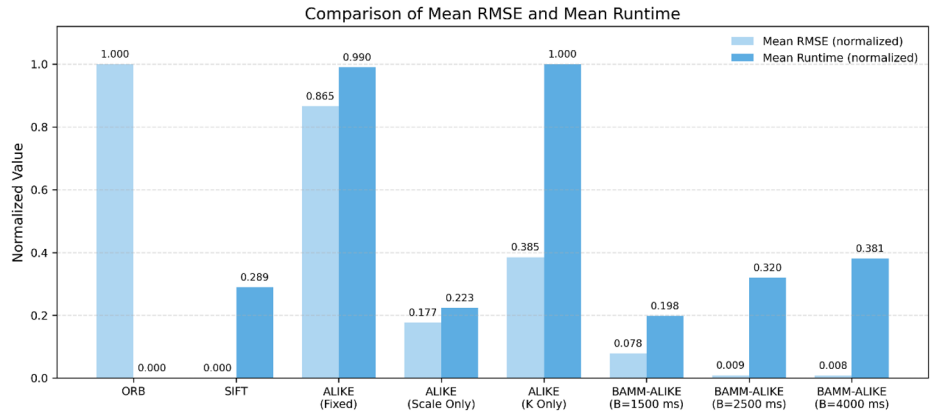


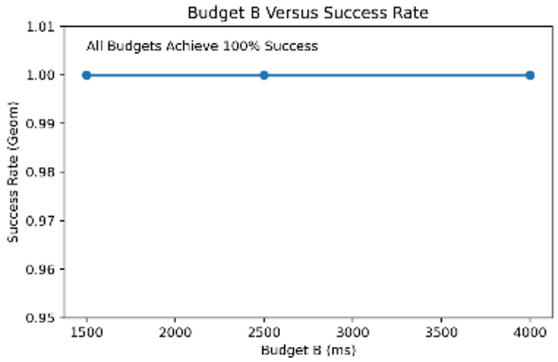
Fig. 8. Mean RMSE of successful samples and average runtime for different methods. (Picture credit: original)

Based on the success rate, ORB and SIFT achieve 0.8000 and 0.8667, respectively. The fixed-parameter ALIKE_fixed reaches 0.7667. This shows that a fixed configuration cannot stably cover all samples. In contrast, ALIKE_scale_only raises the success rate to 1.0000.

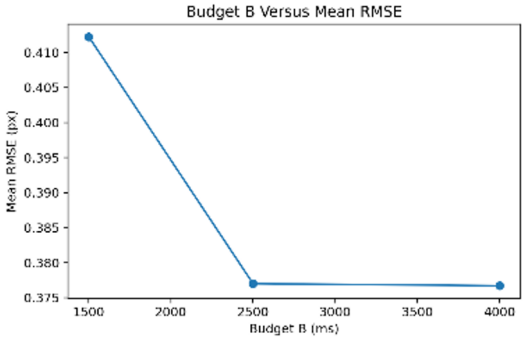
From the perspectives of accuracy and computational cost, SIFT yields a mean RMSE of 0.3727 px on successful samples, but its mean runtime is 833.18 ms. ORB uses only 138.72 ms on average, but its mean RMSE is 0.8779 px. ALIKE_scale_only keeps the success rate at 1.0000. At the same time, it limits the mean RMSE of successful samples to 0.4619 px, and its mean runtime is 676.23 ms. Overall, it achieves a higher success rate, a lower RMSE, and a shorter runtime than fixed ALIKE.

3.2.2 Marginal gain analysis of BAMM

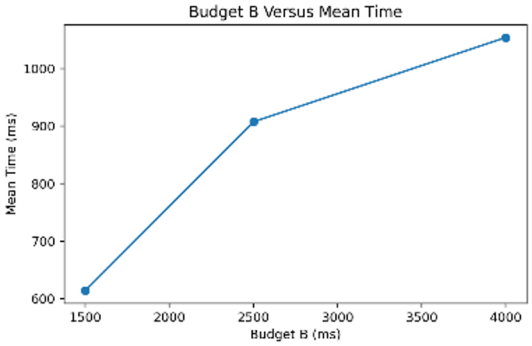
BAMM maintains a success rate of 1.0000 across three computational-budget levels: 1500, 2500, and 4000 ms. As the budget increases, the mean RMSE decreases from 0.4123 px to 0.3770 px. At the same time, the mean runtime increases from 614.15 ms to 1054.17 ms.



(a) BAMM success rate under different computational budgets



(b) Mean RMSE of BAMM under different computational budgets



(c) Mean runtime of BAMM under different computational budgets

Fig. 9. Results of the marginal gain analysis of BAMM (Picture credit: original)

4 Discussion

This study aims to improve the engineering usability of learning-based matching, with ALIKE as the front-end, under a clear time budget. The goal is to achieve stable success and

a controllable trade-off between accuracy and cost. The corresponding hypothesis is as follows: Matching success is not determined only by the number of geometric-verification iterations. The quality of candidate matches and threshold settings more strongly constrain it. Scheduling processing of scale and keypoint resources within a limited computational budget can improve stability. Relaxing the budget can further improve accuracy, but the gain becomes smaller.

In the experiment on the effects of geometric verification parameters, Fig. 4 shows that the control group uses an extremely large sampling upper limit, but its success rate does not improve. This indicates that increasing the number of iterations is not sufficient to convert failed samples into successful ones on this dataset. At the same time, Fig. 5 and Fig. 6 show that changing the threshold affects both the inlier ratio and the error distribution. This forms an observable trade-off between success rate and geometric accuracy. These results indicate that, in engineering practice, simply increasing the upper limit of sampling cannot guarantee stable matching. The quality of candidate matches and the threshold setting must be considered together.

In the multi-method comparison experiment, Fig. 7 shows that fixed-configuration ALIKE still fails on some samples, whereas scale-only adaptation raises the success rate to 1.0000. This indicates that scale selection plays a key role in matchability. Fig. 8 and Fig. 9 show that BAMM-ALIKE maintains a success rate of 1.0000 across all three budget levels. When the computational budget increases from 1500 ms to 2500 ms, RMSE decreases clearly. When the budget increases further, the improvement becomes very small. This reflects diminishing returns. In other words, more computation still improves accuracy, but the gain gradually weakens. Therefore, BAMM turns the operating point under a fixed configuration into an engineering scheduling process with controllable computation and predictable benefit.

This study uses a time budget to characterise the computational budget. It uses processing scale and keypoint resources as allocable variables. It also enables closed-loop scheduling by using geometric verification as feedback. In this way, usability is improved without changing the network structure. The limitation is that the number of budget levels remains limited, and the dataset's coverage still needs expansion. Future work can increase the sampling density of budget settings, extend the method to more complex imaging conditions and larger datasets, and further analyse the optimal resource-allocation pattern under different computational budgets.

5 Conclusion

This paper studies the trade-off among success rate, accuracy, and computational cost in lunar image matching. Under a unified geometric verification framework, it systematically compares RANSAC parameters, threshold settings, and different matching strategies. The results show that simply increasing the RANSAC sampling upper limit does not significantly improve the matching success rate. The upper bound of the success rate is more strongly affected by candidate-match quality and threshold settings. In contrast, threshold changes affect success rate, RMSE, and inlier ratio simultaneously. This reveals a clear trade-off between accuracy and stability. On this basis, this paper further proposes the BAMM-ALIKE method. Through adaptive scale matching under a limited computational budget, it improves the stability and adaptability of lunar image matching. It also shows a good overall balance between accuracy and efficiency in the medium-budget range.

The limitations of this paper are that budget levels remain sparse and that the experimental data are primarily limited to Chang'e-3 image pairs. Scene coverage and geometric-condition coverage are therefore limited. Applicability to more complex lunar environments still needs further verification. Future work can test the method on larger-scale, multi-source, and multi-resolution lunar orbital and landing image datasets. It can also refine the sampling density of

budget settings and combine terrain priors, illumination modelling, and stronger geometric constraints to improve generalisation ability and engineering value.

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